

CCNA Lab Practical

Static Routing Configuration and Verification Six-Router Multi-Path Network Topology

Simulation Platform: Cisco Packet Tracer
Topic: IP Addressing, Static Routes, and End-to-End Connectivity
Course: Cisco Certified Network Associate (CCNA)

Devices	6 Routers (Cisco 2621XM), 3 End Devices (PC-PT)
Routing Method	Static Routing
Addressing	Classful subnets (/8, /16, /24) across point-to-point links
Verification	show ip interface brief, show ip route, extended ping tests

1. Lab Objective

The objective of this lab is to design, address, and configure a multi-router IPv4 network using static routing in Cisco Packet Tracer. The topology connects three end-user segments through a core of six routers arranged with redundant paths, requiring careful planning of subnetting and static route configuration on every router so that all three PCs can reach one another end to end.

This lab demonstrates the following core CCNA skills:

- Designing and assigning an IP addressing scheme across multiple point-to-point WAN links
- Configuring router interfaces with static IPv4 addresses
- Configuring static routes on every router to achieve full network reachability
- Verifying interface status using **show ip interface brief**
- Verifying the routing table using **show ip route**
- Validating end-to-end connectivity using **ping** from each end device

2. Network Topology

The topology consists of six Cisco 2621XM routers (R1–R6) and three end devices (PC1, PC2, PC3). R1 connects the PC1 segment into the core; R2 acts as a central branching point connecting to R1, R6 (which serves PC2), and two parallel routers R3 and R4, which both converge on R5. R5 in turn connects to the PC3 segment. The dual path through R3 and R4 provides redundant connectivity between R2 and R5.

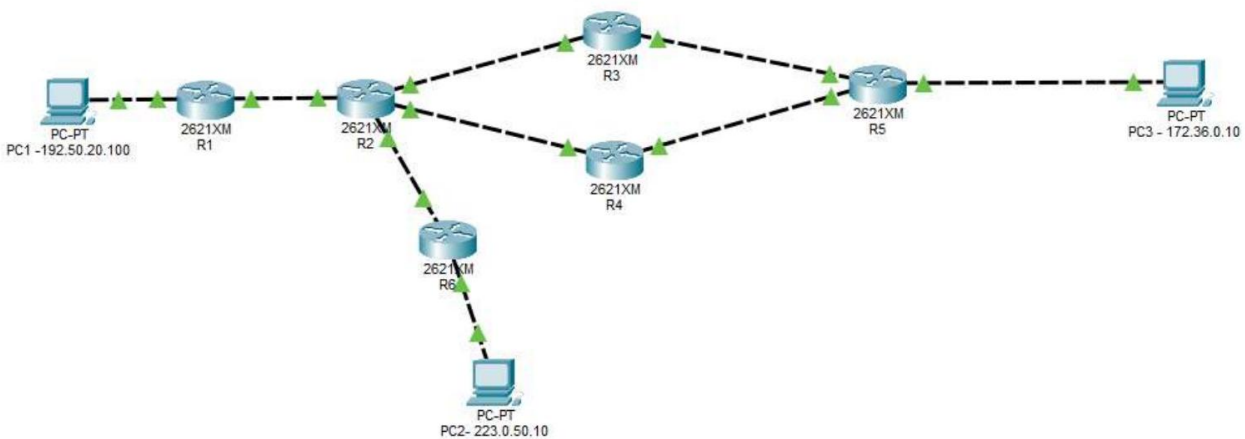


Figure 1 — Logical network topology (device layout)

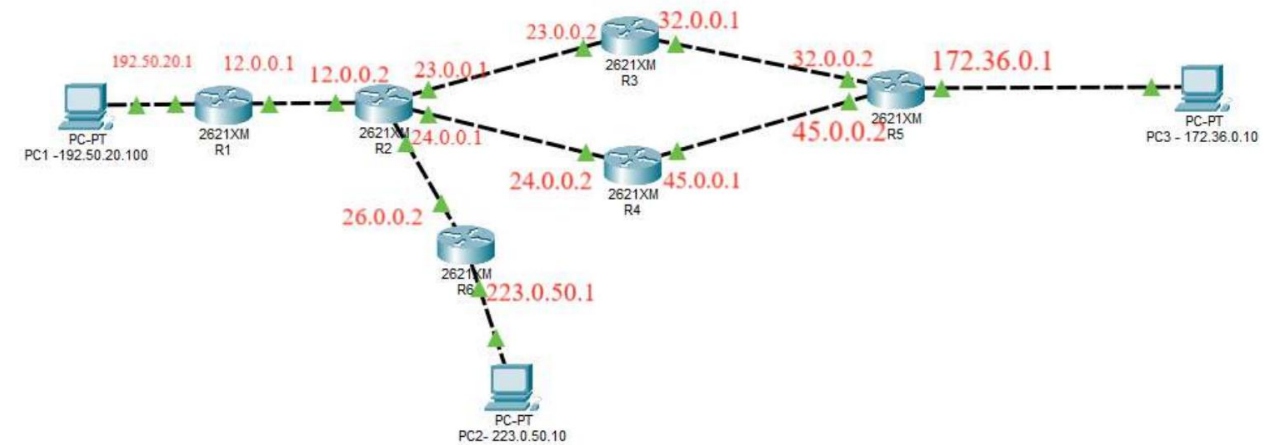


Figure 2 — Network topology with IP addressing scheme applied to every link

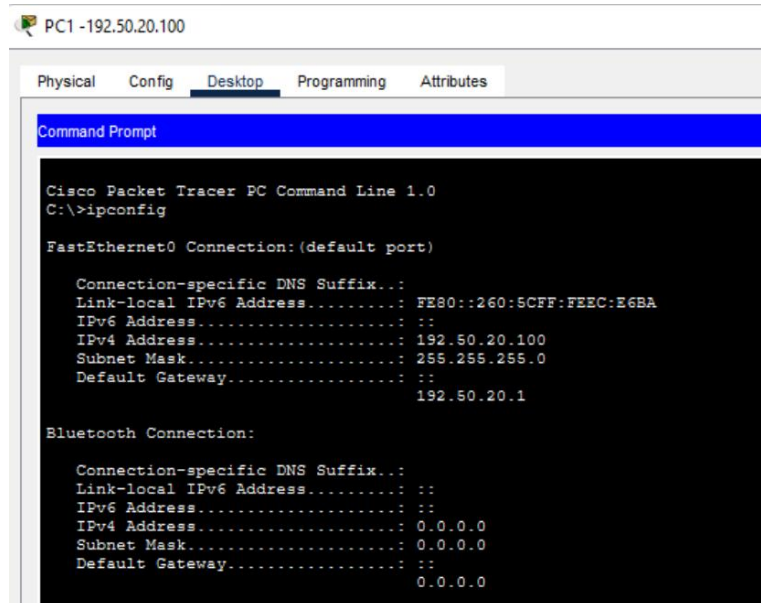
3. IP Addressing Scheme

Each inter-router link and LAN segment was assigned a distinct network as summarized below. End devices use the router-facing interface on their local segment as the default gateway.

Device	Interface	IP Address	Connects To
R1	Fa0/0	192.50.20.1 /24	PC1 (LAN)
R1	Fa0/1	12.0.0.1 /8	R2
R2	Fa0/0	12.0.0.2 /8	R1
R2	Fa0/1	26.0.0.1 /8	R6
R2	Eth1/0	24.0.0.1 /8	R4
R2	Eth1/2	23.0.0.1 /8	R3
R3	Fa0/0	23.0.0.2 /8	R2
R3	Fa0/1	32.0.0.1 /8	R5
R4	Fa0/0	24.0.0.2 /8	R2
R4	Fa0/1	45.0.0.1 /8	R5
R5	Fa0/0	172.36.0.1 /16	PC3 (LAN)
R5	Fa0/1	32.0.0.2 /8	R3
R5	Eth1/2	45.0.0.2 /8	R4
R6	Fa0/0	223.0.50.1 /24	PC2 (LAN)
R6	Fa0/1	26.0.0.2 /8	R2
PC1	FastEthernet0	192.50.20.100 /24	GW 192.50.20.1
PC2	FastEthernet0	223.0.50.10 /24	GW 223.0.50.1
PC3	FastEthernet0	172.36.0.10 /16	GW 172.36.0.1

4. End Device Configuration Verification

The **ipconfig** command was run on each PC to confirm that its IPv4 address, subnet mask, and default gateway were correctly assigned as per the addressing plan.



The screenshot shows the 'Desktop' tab of PC1 (192.50.20.100) in Cisco Packet Tracer. The 'Command Prompt' window displays the output of the 'ipconfig' command. It shows the configuration for the FastEthernet0 interface, including the IPv4 address (192.50.20.100), subnet mask (255.255.255.0), and default gateway (192.50.20.1). It also shows the Bluetooth connection configuration with a default gateway of 0.0.0.0.

```
PC1 - 192.50.20.100

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

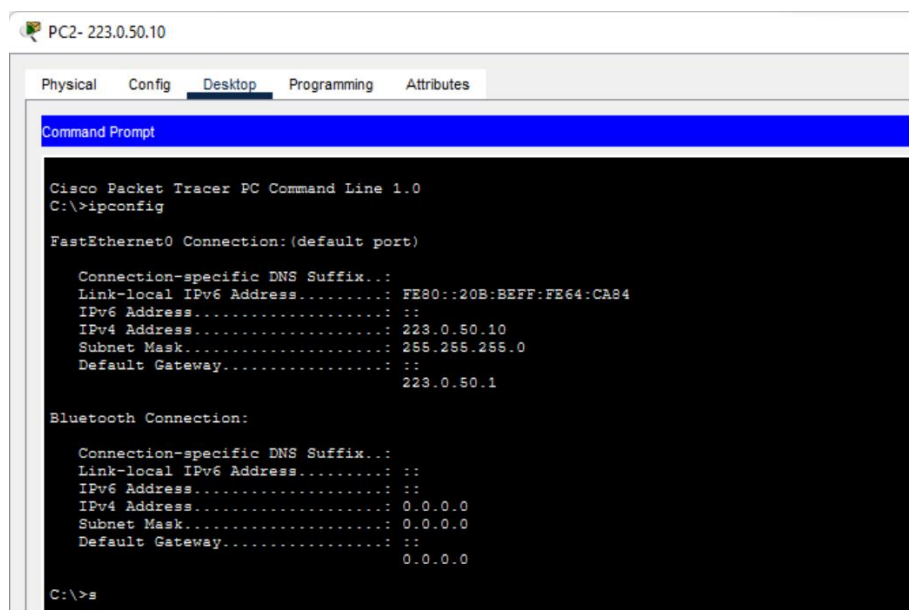
FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: FE80::260:5CFF:FE2C:E6BA
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 192.50.20.100
Subnet Mask . . . . .: 255.255.255.0
Default Gateway . . . . .: ::
                                192.50.20.1

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: ::
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 0.0.0.0
Subnet Mask . . . . .: 0.0.0.0
Default Gateway . . . . .: ::
                                0.0.0.0
```

Figure 3 — PC1 (192.50.20.100) ipconfig output



The screenshot shows the 'Desktop' tab of PC2 (223.0.50.10) in Cisco Packet Tracer. The 'Command Prompt' window displays the output of the 'ipconfig' command. It shows the configuration for the FastEthernet0 interface, including the IPv4 address (223.0.50.10), subnet mask (255.255.255.0), and default gateway (223.0.50.1). It also shows the Bluetooth connection configuration with a default gateway of 0.0.0.0.

```
PC2 - 223.0.50.10

Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

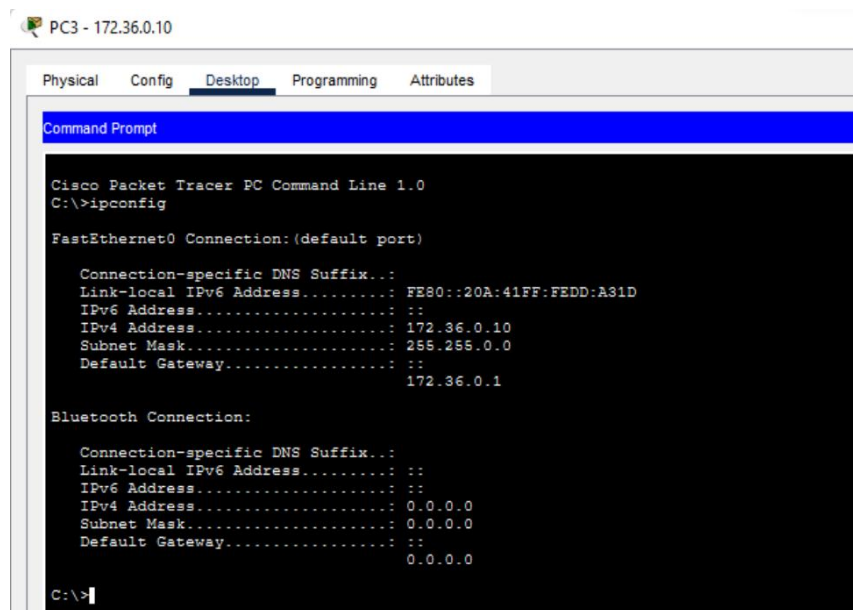
Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: FE80::20B:BEFF:FE64:CA84
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 223.0.50.10
Subnet Mask . . . . .: 255.255.255.0
Default Gateway . . . . .: ::
                                223.0.50.1

Bluetooth Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address . . . . .: ::
IPv6 Address . . . . .: ::
IPv4 Address . . . . .: 0.0.0.0
Subnet Mask . . . . .: 0.0.0.0
Default Gateway . . . . .: ::
                                0.0.0.0

C:\>ipconfig
```

Figure 4 — PC2 (223.0.50.10) ipconfig output



The screenshot shows the Desktop tab of PC3 in Cisco Packet Tracer. The Command Prompt window displays the output of the `ipconfig` command. The output shows the configuration for the FastEthernet0 interface, including the IPv4 address 172.36.0.10, subnet mask 255.255.0.0, and default gateway 172.36.0.1. It also shows the configuration for the Bluetooth interface, which is currently unconfigured (0.0.0.0).

```
PC3 - 172.36.0.10

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20A:41FF:FEDD:A31D
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 172.36.0.10
    Subnet Mask . . . . .: 255.255.0.0
    Default Gateway . . . . .: ::
                                   172.36.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

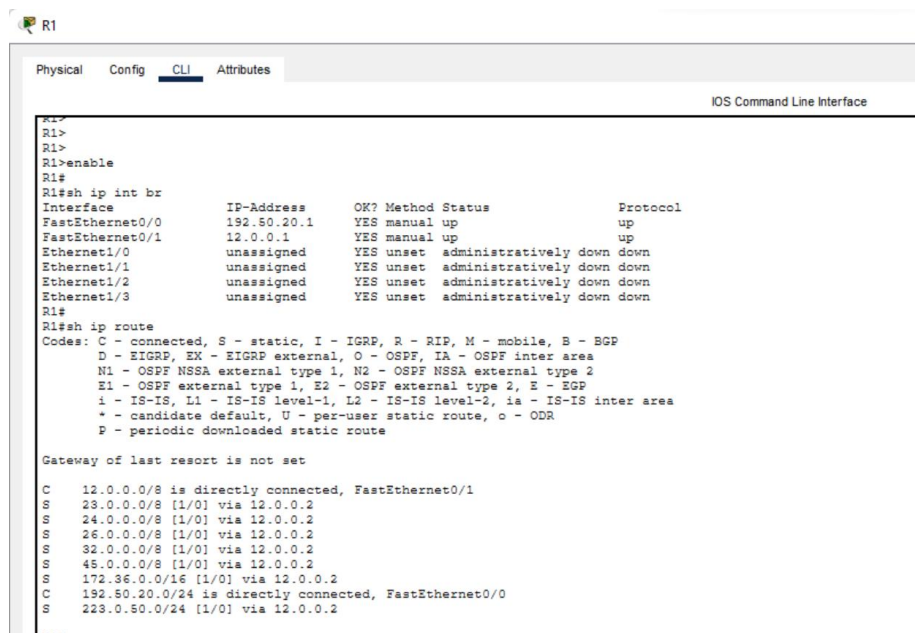
C:\>
```

Figure 5 — PC3 (172.36.0.10) ipconfig output

5. Router Interface and Routing Table Verification

On every router, **show ip interface brief** was used to confirm that all active interfaces were administratively up with a correct IP address, and **show ip route** was used to confirm that static routes (marked **S**) were installed toward every remote network in the topology, in addition to the directly connected (**C**) networks.

5.1 Router R1



```

R1
R1>
R1>enable
R1#
R1#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.50.20.1     YES manual up          up
FastEthernet0/1          12.0.0.1        YES manual up          up
Ethernet1/0              unassigned      YES unset   administratively down down
Ethernet1/1              unassigned      YES unset   administratively down down
Ethernet1/2              unassigned      YES unset   administratively down down
Ethernet1/3              unassigned      YES unset   administratively down down
R1#
R1#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

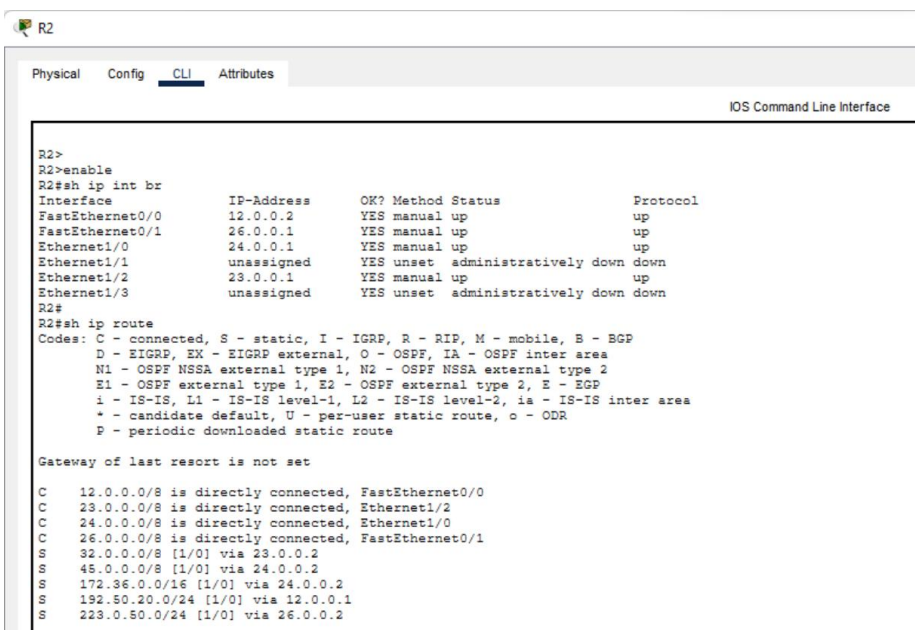
Gateway of last resort is not set

C    12.0.0.0/8 is directly connected, FastEthernet0/1
S    23.0.0.0/8 [1/0] via 12.0.0.2
S    24.0.0.0/8 [1/0] via 12.0.0.2
S    26.0.0.0/8 [1/0] via 12.0.0.2
S    32.0.0.0/8 [1/0] via 12.0.0.2
S    45.0.0.0/8 [1/0] via 12.0.0.2
S    172.36.0.0/16 [1/0] via 12.0.0.2
C    192.50.20.0/24 is directly connected, FastEthernet0/0
S    223.0.50.0/24 [1/0] via 12.0.0.2

```

Figure 6 — R1: interfaces up on Fa0/0 (192.50.20.1) and Fa0/1 (12.0.0.1); static routes to all six remote networks via 12.0.0.2

5.2 Router R2



```

R2
R2>enable
R2#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          12.0.0.2        YES manual up          up
FastEthernet0/1          26.0.0.1        YES manual up          up
Ethernet1/0              24.0.0.1        YES manual up          up
Ethernet1/1              unassigned      YES unset   administratively down down
Ethernet1/2              23.0.0.1        YES manual up          up
Ethernet1/3              unassigned      YES unset   administratively down down
R2#
R2#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    12.0.0.0/8 is directly connected, FastEthernet0/0
C    23.0.0.0/8 is directly connected, Ethernet1/2
C    24.0.0.0/8 is directly connected, Ethernet1/0
C    26.0.0.0/8 is directly connected, FastEthernet0/1
S    32.0.0.0/8 [1/0] via 23.0.0.2
S    45.0.0.0/8 [1/0] via 24.0.0.2
S    172.36.0.0/16 [1/0] via 24.0.0.2
S    192.50.20.0/24 [1/0] via 12.0.0.1
S    223.0.50.0/24 [1/0] via 26.0.0.2

```

Figure 7 — R2: four active interfaces (Fa0/0, Fa0/1, Eth1/0, Eth1/2) reaching R1, R6, R4 and R3; static routes cover the remaining networks

5.3 Router R6



```

R6
-----
Physical  Config  CLI  Attributes

IOS Command Line Interface

LINEPROTO-5-UPDOWN: Line protocol on interface FastEthernet0/0, changed state to up

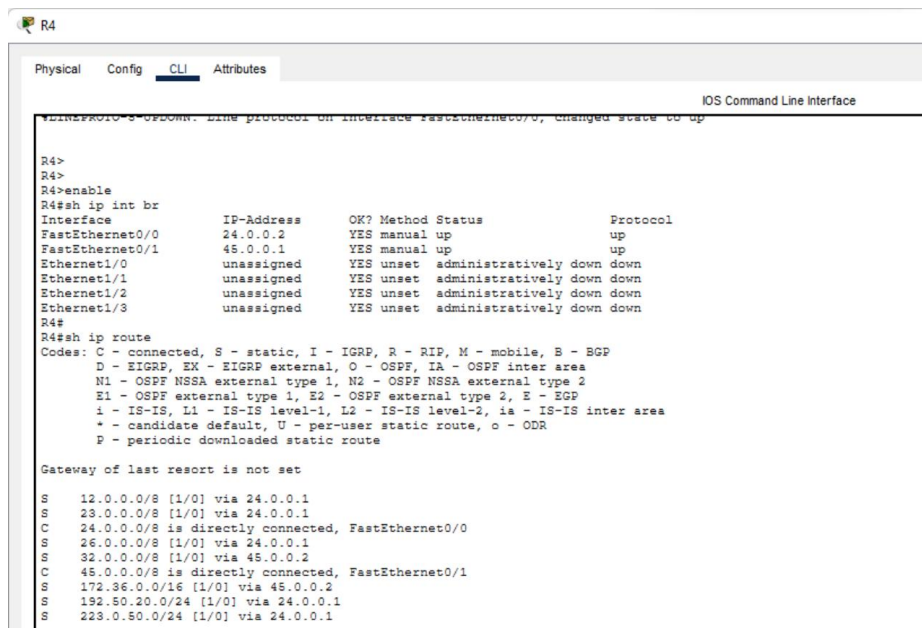
R6>
R6>enable
R6#sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 223.0.50.1      YES manual up          up
FastEthernet0/1 26.0.0.2        YES manual up          up
Ethernet1/0      unassigned      YES unset  administratively down down
Ethernet1/1      unassigned      YES unset  administratively down down
Ethernet1/2      unassigned      YES unset  administratively down down
Ethernet1/3      unassigned      YES unset  administratively down down
R6#
R6#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
C    26.0.0.0/8 is directly connected, FastEthernet0/1
C    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
C    223.0.50.0/24 is directly connected, FastEthernet0/0
  
```

Figure 8 — R6: interfaces up on Fa0/0 (223.0.50.1) and Fa0/1 (26.0.0.2); static default-side routes via 26.0.0.1 toward the rest of the network

5.4 Router R4



```

R4
-----
Physical  Config  CLI  Attributes

IOS Command Line Interface

LINEPROTO-5-UPDOWN: Line protocol on interface FastEthernet0/0, changed state to up

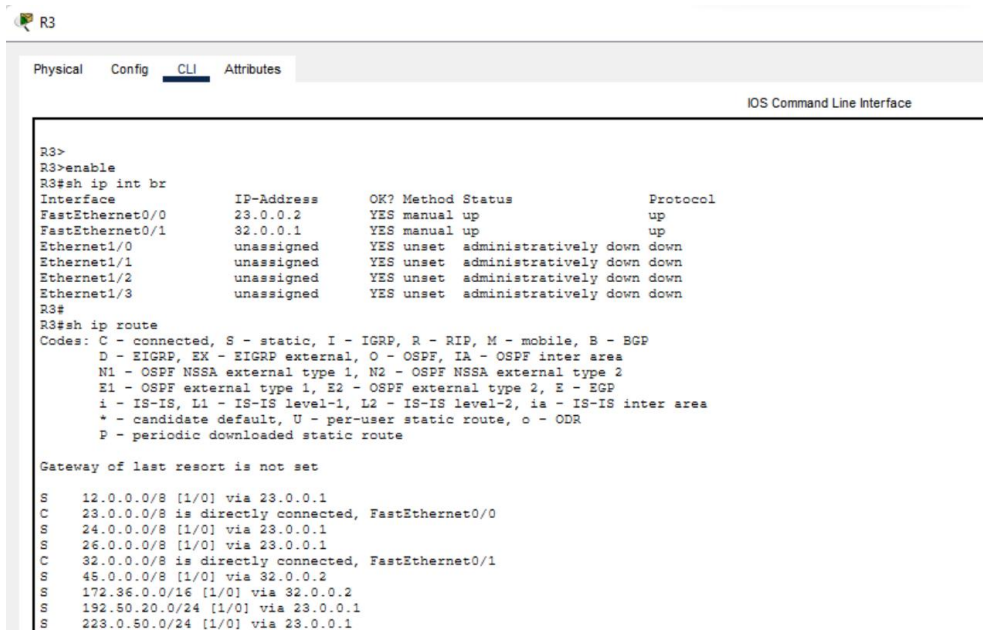
R4>
R4>enable
R4#sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 24.0.0.2        YES manual up          up
FastEthernet0/1 45.0.0.1        YES manual up          up
Ethernet1/0      unassigned      YES unset  administratively down down
Ethernet1/1      unassigned      YES unset  administratively down down
Ethernet1/2      unassigned      YES unset  administratively down down
Ethernet1/3      unassigned      YES unset  administratively down down
R4#
R4#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 24.0.0.1
S    23.0.0.0/8 [1/0] via 24.0.0.1
C    24.0.0.0/8 is directly connected, FastEthernet0/0
S    26.0.0.0/8 [1/0] via 24.0.0.1
S    32.0.0.0/8 [1/0] via 45.0.0.2
C    45.0.0.0/8 is directly connected, FastEthernet0/1
S    172.36.0.0/16 [1/0] via 45.0.0.2
S    192.50.20.0/24 [1/0] via 24.0.0.1
S    223.0.50.0/24 [1/0] via 24.0.0.1
  
```

Figure 9 — R4: interfaces up on Fa0/0 (24.0.0.2) and Fa0/1 (45.0.0.1); static routes via 24.0.0.1 and 45.0.0.2

5.5 Router R3



```

R3>
R3>enable
R3#sh ip int br
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          23.0.0.2        YES manual up            up
FastEthernet0/1          32.0.0.1        YES manual up            up
Ethernet1/0              unassigned      YES unset  administratively down down
Ethernet1/1              unassigned      YES unset  administratively down down
Ethernet1/2              unassigned      YES unset  administratively down down
Ethernet1/3              unassigned      YES unset  administratively down down
R3#
R3#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

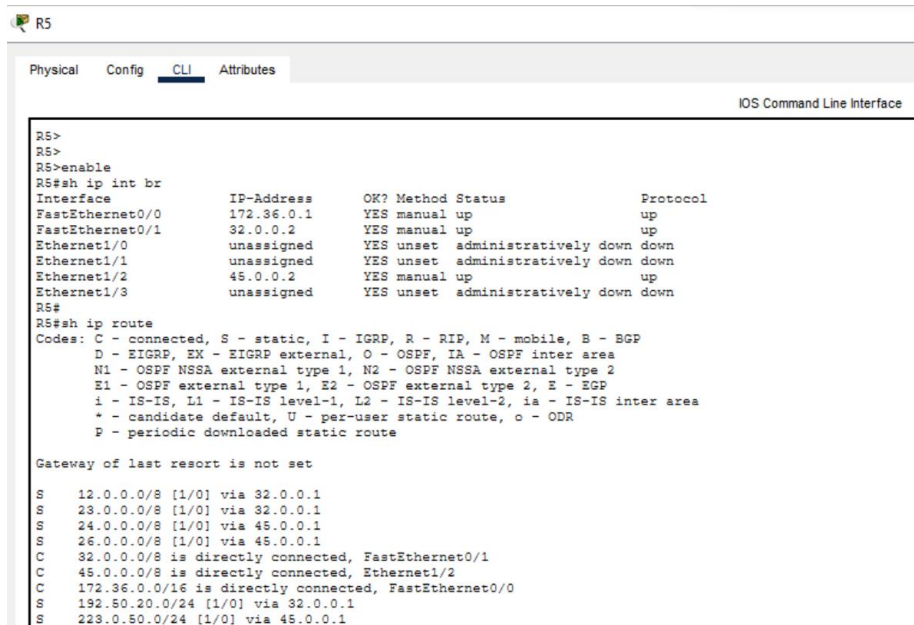
Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 23.0.0.1
C    23.0.0.0/8 is directly connected, FastEthernet0/0
S    24.0.0.0/8 [1/0] via 23.0.0.1
S    26.0.0.0/8 [1/0] via 23.0.0.1
C    32.0.0.0/8 is directly connected, FastEthernet0/1
S    45.0.0.0/8 [1/0] via 32.0.0.2
S    172.36.0.0/16 [1/0] via 32.0.0.2
S    192.50.20.0/24 [1/0] via 23.0.0.1
S    223.0.50.0/24 [1/0] via 23.0.0.1

```

Figure 10 — R3: interfaces up on Fa0/0 (23.0.0.2) and Fa0/1 (32.0.0.1); static routes via 23.0.0.1 and 32.0.0.2

5.6 Router R5



```

R5>
R5>enable
R5#sh ip int br
Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          172.36.0.1      YES manual up            up
FastEthernet0/1          32.0.0.2        YES manual up            up
Ethernet1/0              unassigned      YES unset  administratively down down
Ethernet1/1              unassigned      YES unset  administratively down down
Ethernet1/2              45.0.0.2        YES manual up            up
Ethernet1/3              unassigned      YES unset  administratively down down
R5#
R5#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    12.0.0.0/8 [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
C    32.0.0.0/8 is directly connected, FastEthernet0/1
C    45.0.0.0/8 is directly connected, Ethernet1/2
C    172.36.0.0/16 is directly connected, FastEthernet0/0
S    192.50.20.0/24 [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1

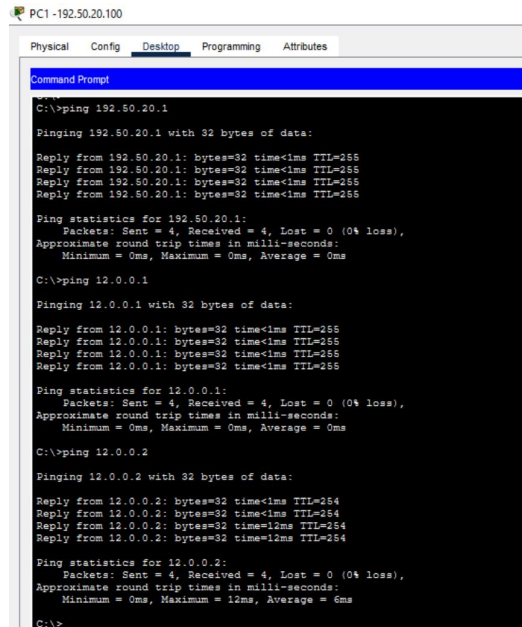
```

Figure 11 — R5: interfaces up on Fa0/0 (172.36.0.1), Fa0/1 (32.0.0.2) and Eth1/2 (45.0.0.2); static routes via 32.0.0.1 and 45.0.0.1

6. End-to-End Connectivity Testing

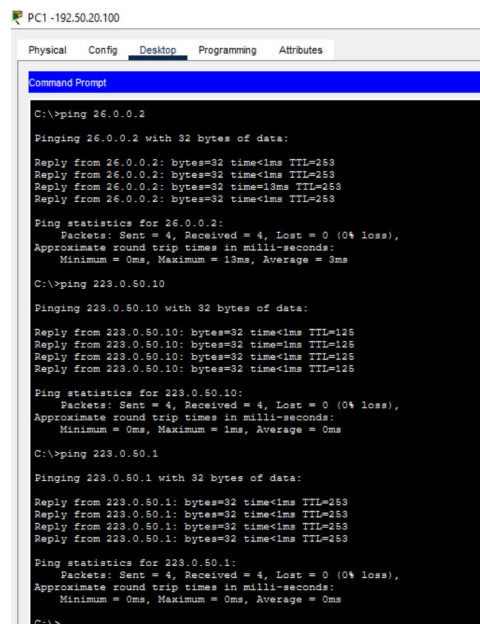
Extended ping tests were run from each PC to every router interface and to the other two PCs. All tests completed with **0% packet loss**, confirming that the static routing configuration provides full, bidirectional reachability across the entire topology.

6.1 From PC1 (192.50.20.100)



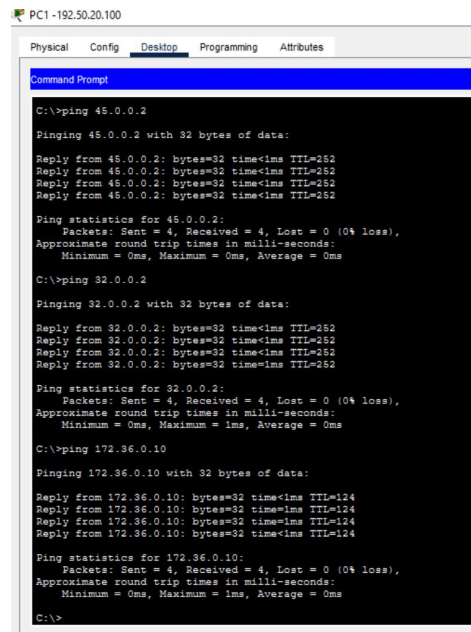
```
PC1 - 192.50.20.100
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 192.50.20.1
Pinging 192.50.20.1 with 32 bytes of data:
Reply from 192.50.20.1: bytes=32 time<1ms TTL=255
Reply from 192.50.20.1: bytes=32 time<1ms TTL=255
Reply from 192.50.20.1: bytes=32 time<1ms TTL=255
Reply from 192.50.20.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.50.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 12.0.0.1
Pinging 12.0.0.1 with 32 bytes of data:
Reply from 12.0.0.1: bytes=32 time<1ms TTL=255
Reply from 12.0.0.1: bytes=32 time<1ms TTL=255
Reply from 12.0.0.1: bytes=32 time<1ms TTL=255
Reply from 12.0.0.1: bytes=32 time<1ms TTL=255
Ping statistics for 12.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 12.0.0.2
Pinging 12.0.0.2 with 32 bytes of data:
Reply from 12.0.0.2: bytes=32 time<1ms TTL=254
Reply from 12.0.0.2: bytes=32 time<1ms TTL=254
Reply from 12.0.0.2: bytes=32 time=12ms TTL=254
Reply from 12.0.0.2: bytes=32 time=12ms TTL=254
Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 6ms
C:\>
```

Figure 12 — PC1 pings its gateway (192.50.20.1) and R2's link interfaces (12.0.0.1, 12.0.0.2) — all successful



```
PC1 - 192.50.20.100
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 26.0.0.2
Pinging 26.0.0.2 with 32 bytes of data:
Reply from 26.0.0.2: bytes=32 time<1ms TTL=253
Reply from 26.0.0.2: bytes=32 time<1ms TTL=253
Reply from 26.0.0.2: bytes=32 time=13ms TTL=253
Reply from 26.0.0.2: bytes=32 time<1ms TTL=253
Ping statistics for 26.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms
C:\>ping 223.0.50.10
Pinging 223.0.50.10 with 32 bytes of data:
Reply from 223.0.50.10: bytes=32 time<1ms TTL=126
Reply from 223.0.50.10: bytes=32 time<1ms TTL=126
Reply from 223.0.50.10: bytes=32 time<1ms TTL=126
Reply from 223.0.50.10: bytes=32 time<1ms TTL=126
Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 223.0.50.1
Pinging 223.0.50.1 with 32 bytes of data:
Reply from 223.0.50.1: bytes=32 time<1ms TTL=253
Reply from 223.0.50.1: bytes=32 time<1ms TTL=253
Reply from 223.0.50.1: bytes=32 time<1ms TTL=253
Reply from 223.0.50.1: bytes=32 time<1ms TTL=253
Ping statistics for 223.0.50.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

Figure 13 — PC1 pings R6's link (26.0.0.2) and reaches PC2 (223.0.50.10) and R6's LAN interface (223.0.50.1) — all successful



PC1-192.50.20.100

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 45.0.0.2

Pinging 45.0.0.2 with 32 bytes of data:

Reply from 45.0.0.2: bytes=32 time<1ms TTL=252
Reply from 45.0.0.2: bytes=32 time<1ms TTL=252
Reply from 45.0.0.2: bytes=32 time<1ms TTL=252
Reply from 45.0.0.2: bytes=32 time<1ms TTL=252

Ping statistics for 45.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 32.0.0.2

Pinging 32.0.0.2 with 32 bytes of data:

Reply from 32.0.0.2: bytes=32 time<1ms TTL=252
Reply from 32.0.0.2: bytes=32 time<1ms TTL=252
Reply from 32.0.0.2: bytes=32 time<1ms TTL=252
Reply from 32.0.0.2: bytes=32 time<1ms TTL=252

Ping statistics for 32.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

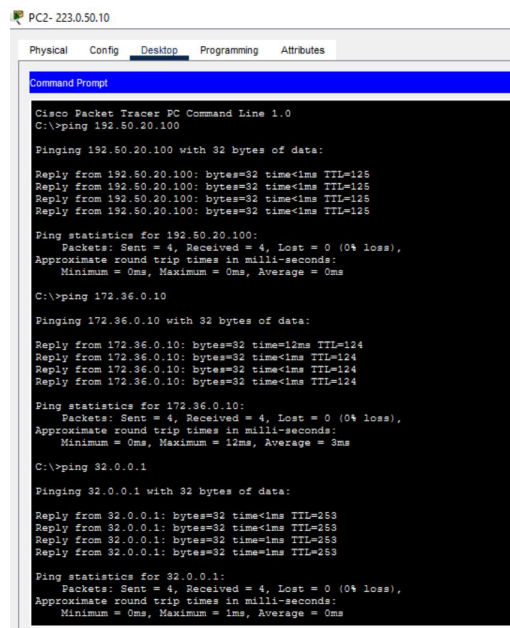
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 14 — PC1 pings R4/R5 links (45.0.0.2, 32.0.0.2) and reaches PC3 (172.36.0.10) — all successful

6.2 From PC2 (223.0.50.10)



```
PC2-223.0.50.10
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=12ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 32.0.0.1

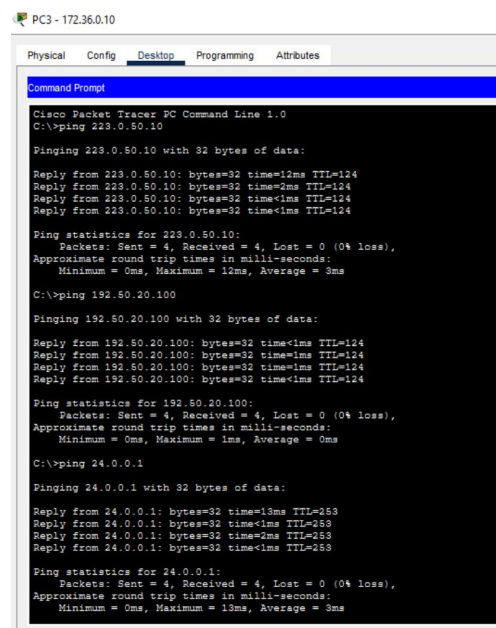
Pinging 32.0.0.1 with 32 bytes of data:

Reply from 32.0.0.1: bytes=32 time<1ms TTL=253
Reply from 32.0.0.1: bytes=32 time<1ms TTL=253
Reply from 32.0.0.1: bytes=32 time<1ms TTL=253
Reply from 32.0.0.1: bytes=32 time<1ms TTL=253

Ping statistics for 32.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 15 — PC2 successfully reaches PC1 (192.50.20.100), PC3 (172.36.0.10) and R3's link interface (32.0.0.1)

6.3 From PC3 (172.36.0.10)



```
PC3 - 172.36.0.10
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=12ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time<1ms TTL=124
Reply from 223.0.50.10: bytes=32 time<1ms TTL=124

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time<1ms TTL=124
Reply from 192.50.20.100: bytes=32 time<1ms TTL=124
Reply from 192.50.20.100: bytes=32 time<1ms TTL=124
Reply from 192.50.20.100: bytes=32 time<1ms TTL=124

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 24.0.0.1

Pinging 24.0.0.1 with 32 bytes of data:

Reply from 24.0.0.1: bytes=32 time=13ms TTL=253
Reply from 24.0.0.1: bytes=32 time<1ms TTL=253
Reply from 24.0.0.1: bytes=32 time=2ms TTL=253
Reply from 24.0.0.1: bytes=32 time<1ms TTL=253

Ping statistics for 24.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

Figure 16 — PC3 successfully reaches PC2 (223.0.50.10), PC1 (192.50.20.100) and R4's link interface (24.0.0.1)

7. Results and Conclusion

All six routers were successfully configured with static IPv4 addressing and static routes covering every remote subnet in the topology. Interface verification confirmed that every link came up correctly, and routing table inspection confirmed that each router held a complete set of routes to all non-local networks. End-to-end ping testing from PC1, PC2, and PC3 confirmed 0% packet loss to every destination in the network, validating that the static routing design achieves full connectivity across the

redundant six-router topology.

Test	Result
Interface status (all 6 routers)	All active interfaces UP / UP
Static routes installed	Complete on all 6 routers
PC1 -> all destinations	0% packet loss
PC2 -> all destinations	0% packet loss
PC3 -> all destinations	0% packet loss